GLADPHARM ASSESSMENT REPORT

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Introduction

Purpose

The purpose of this document is to present the main findings from the assessment conducted. It also defines the project's boundaries across various dimensions, including approach and roadmap, and serves as an agreement between the project team, the project sponsor, and key stakeholders. This document is intended for both business and technical audiences and does not cover the actual system implementation or provide detailed technical guidelines, projections, or presumptions. The Assessment Report covers both business and technical contexts without focusing specifically on either, emphasizing how the technical system implementation benefits the business. This report includes business challenges and recommendations from the analysis phase, and complements the accompanying documentation, deliverables, and diagrams. It is designed to serve as a guide for future teams and resources in implementing Gladpharm future state data platform.

Scope

- Data from multiple sources (ERP, MS SQL and more)
- Main business challenges in data ecosystem
- System Landscape
- Performance requirements and data transfer requirements.
- Users and roles
- Applications Infrastructure/networking components
- Security and Compliance needs
- Recommended changes and improvements
- Proof of Value Use Cases
- Proof of Value Scope
- Deployment Plan

Out of Scope

- Analysis of current business processes.

Definitions

Definitions and Abbreviations

#	Term	Description
1	DWH	Data Warehouse
2	REST	Representational State Transfer
3	API	Application Programming Interface
4	ETL	Extract, Transform, Load
5	ELT	Extract, Load, Transform
6	POV	Proof of Value
7	BI	Business Intelligence
8	BG	Business Goal
9	CI	Continuous Integration
10	CD	Continuous Delivery/Continuous Deployment
11	CSV	Comma-Separated Values
12	GB	Gigabyte
14	SQL	Structured Query Language
15	DB	Database
16	EoM	End of Month
17	EoQ	End of Quarter
18	EoY	End of Year
19	SLA	Service-Level Agreement
20	DEV	Development
21	PROD	Production
22	VLAN	Virtual Local Area Network
23	RBAC	Role Based Access Control
24	HTTPS	HyperText Transfer Protocol Secure
25	VM	Virtual Machine
26	ODS	Operational Data Store

Business overview

Business Background

Gladpharm Ltd is a Ukrainian pharmaceutical manufacturer established in 1996 that has grown into one of the leading domestic players. Its core business involves developing, manufacturing and supplying generic medicines across several therapeutic areas including cardiology, endocrinology and gastroenterology, while adhering to international-GMP standards.

From the business-perspective, Gladpharm places emphasis on building resilient local production capabilities, investing in modern infrastructure and quality-control systems, and positioning itself as a reliable partner in Ukraine’s healthcare ecosystem. The company also shows strong social responsibility and national-service orientation—highlighting its role in the supply of medicines during challenging times in Ukraine’s environment.

Problem statement

The main disadvantages of the existing solution are:

- Part of the company's data processing and reporting workflows is still performed manually, which increases effort and limits the frequency of report updates.
- Data from multiple operational systems is not yet fully integrated, which complicates end-to-end analytics and consolidated reporting.
- The company manages over 44 million records annually across numerous systems (ERP, OData, REST API, Excel, Dataverse, Azure DB, SharePoint, etc.), creating complexity in data handling and synchronization.
- Lack of a unified data platform results in disconnected processes and repeated data preparation efforts.
- Data-quality control is not fully standardized, as validation processes differ between systems.
- The current setup limits scalability, flexibility, and the ability to efficiently produce accurate and timely analytical outputs.

Main Business Challenges and Needs

Key Business Challenges

Key Business Challenges	Business Perspective	Technical Perspective
1. Decentralized data architecture and limited flexibility	Dispersed data across numerous operational systems leads to duplicated efforts and higher maintenance costs.	Absence of a centralized data platform and governance model limits integration, scalability, and agility in developing new analytical solutions.
2. Manual and time-consuming reporting processes	Manual preparation of summary tables increases workload and slows down reporting cycles, limiting the ability to update reports frequently and make timely business decisions.	The existing internally developed data-processing solution lacks full automation and central orchestration, which reduces efficiency and complicates maintenance across multiple data sources.
3. Inconsistent data quality and limited governance	Incomplete or inconsistent data across reports reduces trust in analytics and decision-making accuracy.	Data-quality checks are applied unevenly, with no unified or automated validation framework implemented across all systems.

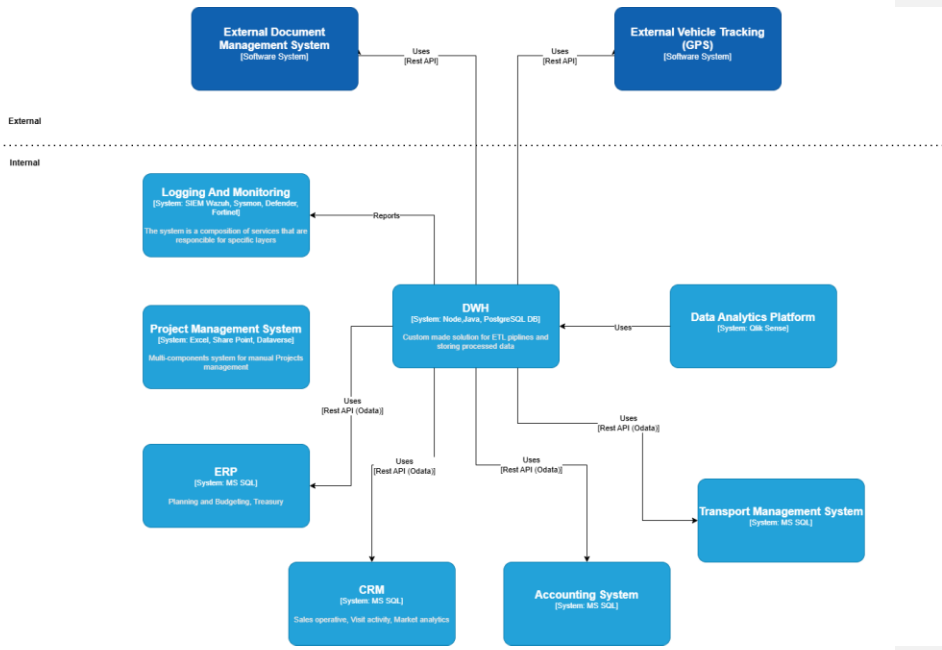
Business Needs and use cases

Business Needs	Description
1. Ensure Business Continuity	Implement a robust and resilient data platform capable of processing high workloads from multiple operational systems while maintaining stable performance and ensuring uninterrupted analytical and reporting activities.
2. Efficient Analytical Reporting	Develop an automated reporting environment that minimizes manual effort, enables daily or on-demand report generation, and supports timely and accurate business decision-making.
3. Scalable and Integrated Solution	Establish a scalable, centralized architecture that consolidates data from all key sources (ERP, finance etc.) and ensures consistent, high-performance data access for all divisions.
4. Enhanced Data Quality	Improve overall data quality by standardizing validation and control mechanisms, automating checks, and implementing unified governance procedures across the organization.

Applications Infrastructure

System Landscape

Gladpharm operates a diverse ecosystem of internal and external systems. Most systems are internal, connected via REST API or OData interfaces, while several external services (document management, GPS tracking) exchange data through REST APIs. Data is primarily stored in MS SQL databases and complemented by Excel, Dataverse, Azure DB, and SharePoint for additional inputs and collaboration. Analytical reporting is supported through Qlik Sense, which consumes and visualizes data from these sources.



Systems and services

#	System	Description
1	OData-based systems	Internal business applications (such as ERP, planning) that provide access to operational and financial data through OData REST APIs within the internal infrastructure.
2	BAF	Accounting and transport management systems that provide access to operational and financial data through OData REST APIs within the internal infrastructure.
3	MS SQL	Microsoft SQL Server includes several key data domains, which are accessed directly within the internal network or through API connections for analytical and reporting purposes.
4	Systems exposing data via	A set of internal and external systems, including Azure DB and Share Point, that exchange data through REST API endpoints.

	RESTful APIs	These integrations enable automated data retrieval, synchronization, and interaction with external services.
5	Excel	Used as an internal data source for manual input, local data storage, and departmental reporting. Excel files are shared within the organization and can be integrated into analytical workflows through connectors or automated import processes.
6	Dataverse	A cloud-based data service used for structured data storage and integration with other business applications. Data is available through the OData Web API, enabling consistent access and automation within internal business processes.
7	Qlik Sense	A business intelligence and data visualization platform used for building dashboards and analytical reports. It allows users to connect to various data sources, perform transformations, and create interactive visualizations for business analysis.

Data Processing

Data processing within Gladpharm is currently performed through an internally developed integration platform. This system manages ETL/ELT workflows — data extraction, transformation, and loading — and supports horizontal scalability to handle growing data volumes.
Processed data is then loaded into analytical environments such as Qlik Sense, where it is used for reporting, performance monitoring, and management dashboards.

Data Migration needs

Migrating data from an existing solution, especially when it comes to historical data, requires careful planning and several specific requirements to ensure a smooth, secure, and successful process.
Here are the key requirements for this process:

- Database size and complexity: Estimate the size of the database and the complexity of its schema, including tables, indexes, stored procedures, triggers, and functions.
- Performance requirements: Assess the current performance.
- Initial data loading: Perform an initial bulk transfer of historical data to the cloud environment.

- Incremental data upload: If required, use incremental data upload techniques to synchronize changes made during the initial upload phase.
- Verify data: Validate the migrated data to ensure that it matches the source data in terms of integrity and consistency.

Performance requirements and data transfer requirements.

Data Processing Schedule

Data Source	Processing Deadline	Average Processing Time	Maximum Processing Time
Raw Data (ERP, external)	05:00 AM	2 hour	3 hours
Dimensions Data	5:30 AM	15 min	1 hour
Fact Tables	7:00 AM	30 minutes	1 hour
Data Marts	8:00 AM	20 minutes	40 minutes

Incoming data volumes per source

Source	Daily data volume, MB	Data type	Location
ERP	6	Structured data exposed via OData protocol	Internal server
Accounting System	2	Structured data exposed via OData protocol	Internal server
Transport Management System	3	Structured data exposed via OData protocol	Internal server
MS SQL	60	Relational data (MS SQL)	Internal server

Excel (Project Activity)	0.1	Flat files (Excel / CSV)	Local computers
Azure DB (Document Management)	0.5	Semi-structured data (JSON from REST API)	External
Vehicle Tracking (GPS)	0.5	Semi-structured data (JSON from REST API)	External
Other Sources (SharePoint, Dataverse)	0.3	Semi-structured data	Internal server

Performance and SLA Considerations

Performance Metric	Requirement
Data processing during peak times (EoM/EoQ/EoY)	Should not impact existing SLAs; prioritize new data
Response time for 80% of standard queries	Within 2 minutes
Impact of ad-hoc queries on solution performance	None (should not affect data processing, refreshes, etc.)

The data presented above are approximate estimates based on currently available information. Actual volumes may vary depending on the reporting period, business activity, and integration frequency. These figures are intended for initial planning and assessment purposes and can be further refined.

Users and roles

#	Role	Description
1	Administrator	Manages infrastructure, user access, and security settings.

2	Support	Monitors solutions, handles deployments, and ETL/ELT processes. Focuses on PROD environment with read-only access to address user questions and issues.
3	Power BI Developer	Develops reports and dashboards with read-only access.
4	Developer	Manages development processes with full control over the DEV environment. Can modify the solution codebase for each component if necessary. Has read-only access to PROD environment.
5	DevOps	Automates deployment processes. Implements IaC principles using Terraform HCL.
6	Data Owner	Manages data, including access permissions, usage policies, and monitoring data usage.
7	Data Analyst	Handles data analysis processes, develops new data marts, and optimizes existing processes.

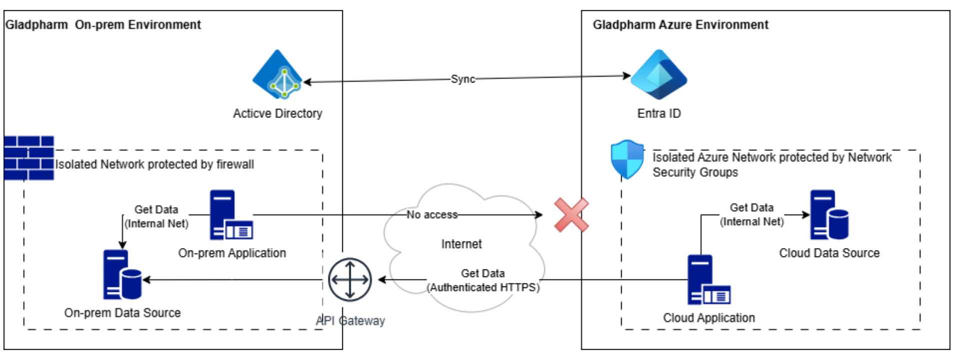
	Role	Members
1	Administrator	TBD
2	Support	TBD
3	Power BI Develop	TBD
4	Developer	TBD
5	DevOps	TBD
6	Data Owner	khomenko.r@gladpharm.com
7	Data Analyst	khomenko.r@gladpharm.com

Data classification and related risks

Data Source	Description	Classification	Risk
ERP	Stores core enterprise data related to financial planning, budgeting, treasury operations, and internal cost management.	Restricted	High risk — contains sensitive financial information; unauthorized access could result in financial loss or compliance violations.
Accounting System	Holds accounting records, general ledger entries, and financial reporting information and management accounting.	Restricted	High risk — includes confidential financial data; exposure could cause financial misstatements or regulatory non-compliance.
Transport Management System	Contains operational data on fleet management, vehicle usage, routes, and logistics performance.	Internal Use Only	Medium risk — may expose internal logistics operations and transport schedules.
MS SQL	Includes data domains: Operational Sales (sales transactions and performance data), Visit Activity (information about promotional interactions of medical representatives with healthcare professionals, conferences, and digital communication), and Market Analytics (data on market trends and competitive positioning).	Confidential	High risk — includes personally identifiable and commercial data; unauthorized access may cause privacy breaches, reputational harm, or competitive disadvantage.
Excel	Contains project management data, including task tracking, timelines, deliverables, and resource allocation information.	Internal Use Only	Medium risk — exposure could reveal internal project structures, delivery timelines, or confidential development information.
Azure DB	Stores business documents, contracts, and approval workflows exchanged internally and with external partners.	Internal Use Only	Medium risk — unauthorized access could expose internal or partner

			documentation and agreements.
Vehicle Tracking (GPS)	Holds data on vehicle locations, movement history.	Internal Use Only	Medium risk — exposure may reveal logistical patterns, transport routes, or internal mobility details.
Other sources (Share Point and Dataverse)	Contain semi-structured and reference data used for collaboration, document storage, and cross-departmental integrations.	Internal Use Only	Medium risk — exposure may reveal internal documents, shared resources, or integration data used in operational processes.

Network Architecture



The customer's infrastructure is a hybrid IT system, hosted within:

- Local data centers and secured by Fortinet network appliances, which provide perimeter protection, traffic filtering, and basic intrusion prevention capabilities.
- Azure Environment with isolated cloud network protected by Azure services such as Network Security Groups.

There is currently no dedicated private channel to Azure such as ExpressRoute or VPN Gateway, meaning that all communication with external or cloud services occurs over the public internet using standard secure protocols (HTTPS, TLS). IP Allow lists and Obligatory Authentication provides additional layer of defense.

Network and endpoint monitoring are handled through Fortinet ecosystem, which serves as the central network management system for on-prem Data Center.

- The organization gathers a comprehensive set of logs, including:
- Security agent logs and system events from endpoints.
- Sysmon data such as process launches, file creations, and DNS queries.
- Audit logs from Microsoft Defender.
- Fortinet network traffic logs, capturing ingress and egress flows.

At present, the environment does not utilize Azure Monitor or Log Analytics, and there is no centralized telemetry pipeline covering both on-premises and cloud resources. Monitoring is therefore siloed, focusing mainly on infrastructure-level and security events rather than application-level observability.

Connectivity from cloud application to on-prem data sources is established through authenticated public endpoints served by custom API Gateway instances. Identity management and access control is handled on a per-system basis, with limited integration into a unified identity provider such as Azure AD.

Both on-prem Active Directory and Azure Entra ID a configured to perform synchronization.

Security and Compliance needs

Security Measure	Explanation
Data Encryption	Implement encryption at-rest to protect sensitive data, such as actual income~~.~~ This ensures that even if unauthorized access occurs, the data remains unreadable and unusable, safeguarding its confidentiality and integrity.financial indicators.
Access Control	Utilize access control mechanisms to restrict access to the analytical solution and its underlying data. Implement role-based access control (RBAC) to ensure that only authorized personnel have access to specific data and functionalities based on their roles and responsibilities.

Secure Communication	Employ security protocols (e.g., HTTPS) to encrypt data transmission between different components of the analytical solution, including sources, integration components, data warehouse, analysis tools, and user interfaces. This prevents data tampering during data exchange, enhancing data confidentiality and integrity.
Geography Optimization	Optimize the solution for minimal latency for Ukrainian users, considering that all end users reside in Ukraine. Consider regional planning to address data availability and other region-specific risks, ensuring optimal performance and mitigating potential challenges.
Data Integrity Checks	Implement mechanisms to ensure data integrity throughout the data processing pipeline. Utilize hash functions to verify the integrity of data at different stages of processing, detecting any unauthorized alterations and maintaining data accuracy and reliability.

By integrating these security measures into the solution, the company can mitigate security risks, protect valuable data assets, and build trust with stakeholders. This enhances the resilience and reliability of its operations, ensuring the confidentiality, integrity, and availability of sensitive data.

Security analytical solution is crucial to protect sensitive data, maintain integrity of operations and safeguard against potential threats, The following list of security measures integrated into the solution:

1. Data Encryption: Implemented encryption at-rest to protect sensitive data, such as financial indicators.
2. Access Control: Access control mechanism restricts access to the analytical solution and its underlying data. Implemented role-based access control (RBAC) ensures that only authorized personnel have access to specific data and functionalities based on their roles and responsibilities within the company.
3. Secure Communication (encryption in-transit): Security protocols (HTTPS) should be employed to encrypt data transmission between different components of analytical solution, including sources, integration components, data warehouse, analysis tool and user interfaces. This prevents from data tampering during data exchange.

4. Geography: All end users that will use the solution resided in Ukraine. Global In this case global regions are redundant, and the solution should be optimized to have minimal latency for Ukrainian's users. Also, data availability and other region risks considered in the regional planning.
5. Data Integrity Checks: Implement mechanisms to ensure data integrity through the data processing pipeline. Hash functions utilized to verify integrity of data at different stages of processing.

Integrating these security measures into the solution the company can mitigate security risks, protect valuable data assets and build trust with stakeholders, enhancing the resilience and reliability of its operations.

Resilience, Availability, and disaster recovery requirements

Customer Expectations	Solution Requirements
Expected Demand and Scalability	- Solution should absorb peak times, accommodating data volume growth up to 75% of baseline. - Peak times typically occur at month, quarter, and year ends. - Integrated monitoring system needed for scaling up components. - Data warehouse database and ETL components must support elastic scaling without recreation and minimal downtime. - Raw data stored in low-cost storage for analysis and re-processing.
Uptime and SLAs	- Solution uptime: ≥96%. - Data warehouse SLA on weekdays: ≥98%. - Maintenance window: Saturday 4 pm to Sunday 4 pm to avoid business hour disruptions.
Backup and Recovery	- Data stored on storage replicated to ≥2 data centers. - Automatic daily backups for data warehouse database. - Backup files replicated to ≥2 data centers. - Point-in-time recovery supported. - Full restore within 24 hours. - Automatic backup configured for all VMs. - Centralized configuration storage for cloud components, including change history.

Scalability

One of the main reasons for migrating current solutions to the cloud is scalability. The customer expectations are:

- The solution should absorb peak time when data volume can grow to 75% of the baseline. The usual peak time for the customer is EoM, EoQ and EoY.
- The customer wants to have an integrated monitoring system that advises when scale-up components are reqsuired
- Data warehouse database, ETL components should support elastic scaling up and down without re-creation and with minimal downtime
- Network connectivity between on-premises network and Azure is out of scope.
- Row data received from source systems should be stored in low-cost storage with the ability to analyze and re-process this data without access to the corresponding source system.

Backup and Recovery

- Data stored in storage should be replicated to at least two different data centers.
- Data warehouse database should have automatic backup enabled and make copies daily
- Data warehouse backup files should be replicated to at least two different data centers.
- Restore from backup should support point-in-time recovery
- Full recovery should take no more than 24 hours
- All virtual machines containing data should have automatic backup configured
- All cloud component configurations should be centralized and stored separately (including change history).

Component SLA

- Solution should have at least 96% SLA
- Data warehouse should support SLA at least 98% during weekdays
- Maintenance window from Saturday 16:00 to Sunday 16:00

Recommended changes and improvements

This document chapter covers the main opportunities and directions where the Kyivstar team sees the most valuable potential for future changes in Gladpharm.

#	Component	General Description	Detailed Description
1	Centralized Data Platform	Establish a unified data environment for all operational and analytical needs	Implement a consolidated platform that integrates data from ERP, MS SQL, Accounting, Transport, and Project systems. This will serve as a single source of truth, simplifying access, improving data reliability, and supporting analytics and reporting at the enterprise level.
2	Data Quality and Governance Framework	Define policies and automation for consistent, reliable data	Introduce standardized data quality rules, validation checks, and governance policies across all data domains. Automating these controls will minimize manual intervention and ensure compliance with data management standards.
3	Templating the manual data sources	Set of policies and processes to control the manual data sources	The manual data sources are closely related to the data producers, which are humans, who can make a lot of mistakes in the data creation processes. To reduce the chance to make a mistake it's a good practice to create the templates for manual files (for example templates of Excel files, which contains manual references)
4	CI/CD tool	The CI/CD implementation tool	CI/CD is a very important part of the development process. The CI/CD implementation tool allows to control codebase versioning, automate the code deployment across the different environments. It is also a single source of code, which can be shared between the multiple developers

The recommended changes and improvements above aim to improve and optimize the Gladpharm existing workflows during the implementation of the new data consolidation platform. By implementing them Gladpharm can use its data more effectively to get more business value.

Use Cases

Considering the existing business challenges the next use cases aligned with business goals should be addressed

UC-ID	UC description	UC benefits	Priority

UC-01	Enterprise Data Consolidation Platform	Build a unified data consolidation platform that integrates all operational systems (ERP, MS SQL, Finance, Treasury, Logistics, Projects, etc.) into a single, governed environment. This will provide a single source of truth, enable cross-domain analytics, and reduce duplicated data processing across departments.	1
UC-02	End-to-End Automation and Scalability	Implement automated data workflows and scalable pipelines to process large data volumes efficiently and reliably. This reduces manual interventions and improves performance during peak data processing periods.	1
UC-03	Advanced Analytical Capabilities	Develop advanced dashboards and analytics powered by high-quality, consolidated data. Automation within Microsoft Fabric reduces the number of manually built reports and provides faster, insight-driven decision-making.	1
UC-04	Data Governance and Quality Management	Introduce a standardized governance framework covering data validation, lineage, access control, and compliance. This will ensure transparency, accuracy, and regulatory alignment across all data assets.	2
UC-05	Operational Efficiency and Cost Optimization	Optimize infrastructure and data-processing resource utilization through scalable architecture, reducing total cost of ownership while maintaining performance.	2
UC-06	Improved User Experience and Collaboration	Provide business users and analysts with a modern, accessible data environment that simplifies collaboration, accelerates report creation, and ensures consistent use of reliable data.	2

Proof of Value (POV) Use Case Scope

Description

The Use Case will reproduce the existing Excel report “Бюджет наступного року по ЦФВ” within Microsoft Fabric and Power BI. Data from ERP used for budgeting and financial reporting, along with reference catalogs (budget items, cost centers, and organizational units), will be ingested, transformed, and aggregated automatically, replacing manual data preparation and consolidation.

The resulting Power BI dashboard will replicate the structure of the current Excel report, allowing filtering by organization, CFR, cost center, employee, and period, with export options for management review.

Scope

- Data sources: ERP database
- Output: Power BI budget report with an agreed structure
- Processes tested: data ingestion, transformation, reporting
- Evaluated: data accuracy, report responsiveness, refresh speed, and user interactivity

Expected Outcomes

- Demonstrated automation of previously manual Excel-based budgeting and reporting workflows
- Reduced manual effort through automated data ingestion, transformation, and refresh processes in Fabric
- Improved data accuracy and consistency by consolidating data in a single environment
- Showing how Microsoft Fabric simplifies reporting, ensures data integrity, and provides a scalable foundation for future analytics at Gladpharm

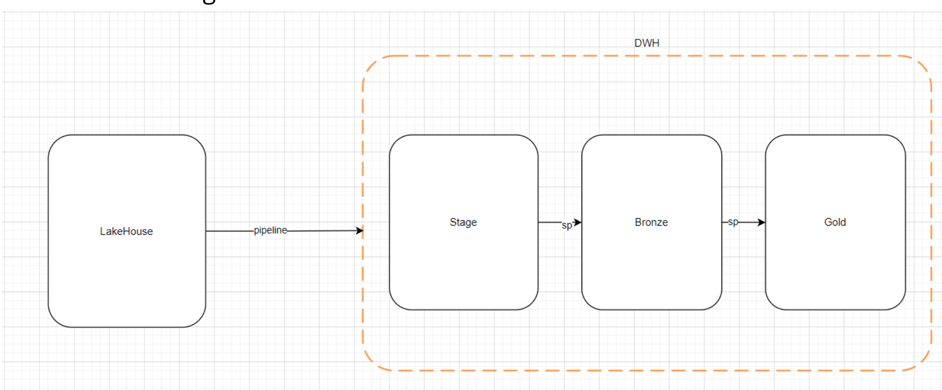
Success Criteria

The Proof of Value will be considered successful if Microsoft Fabric enables the full automation of the report previously created manually in Excel, daily or on-demand refresh through Fabric pipelines, and delivery of an interactive Power BI report that replicates the existing structure while significantly reducing manual effort and demonstrating readiness for scalable, governed analytics within Gladpharm.

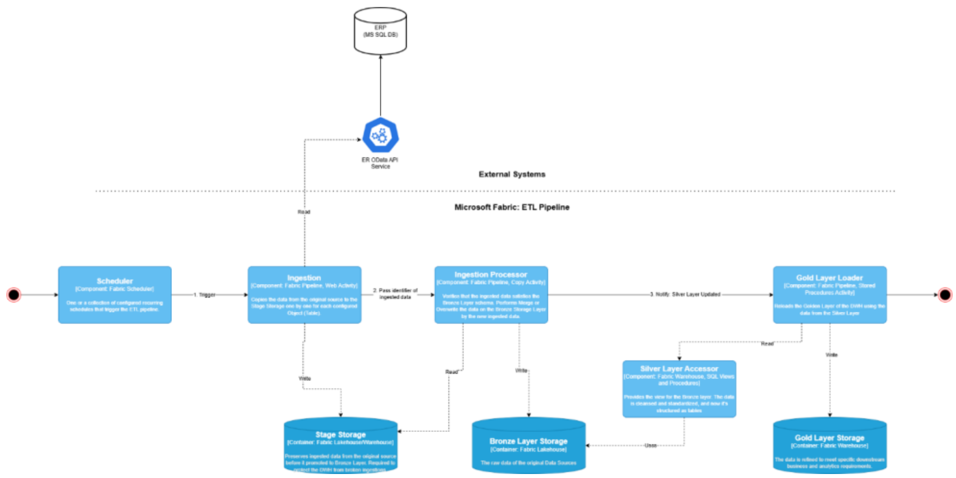
Data Processing

- The source data is stored with monthly granularity
- Data from ERP instance will be transmitted through a REST API using OData protocol.
- Time table of data collection: 1 time per day from January to September. 1 time per hour – from October to December.
- Perform a Full Reload of the DWH.

Structure of Storage



The whole ETL flow is described on the following diagram:



As an initial requirement business expects to see following objects to be imported:

- Статьи Бюджетов
- Статьи Оперативных Бюджетов
- Статьи ОБ_СтатьиБюджета
- Пользователи
- Структура Предприятия
- Организация
- Каз_БюдАдминВитрат_
- ОБ_АдмінВитрат_RecordType

*** First 6 objects are Dimensions and next 2 are Facts

These objects are stored on **Lakehouse** as-is with following the naming convention:

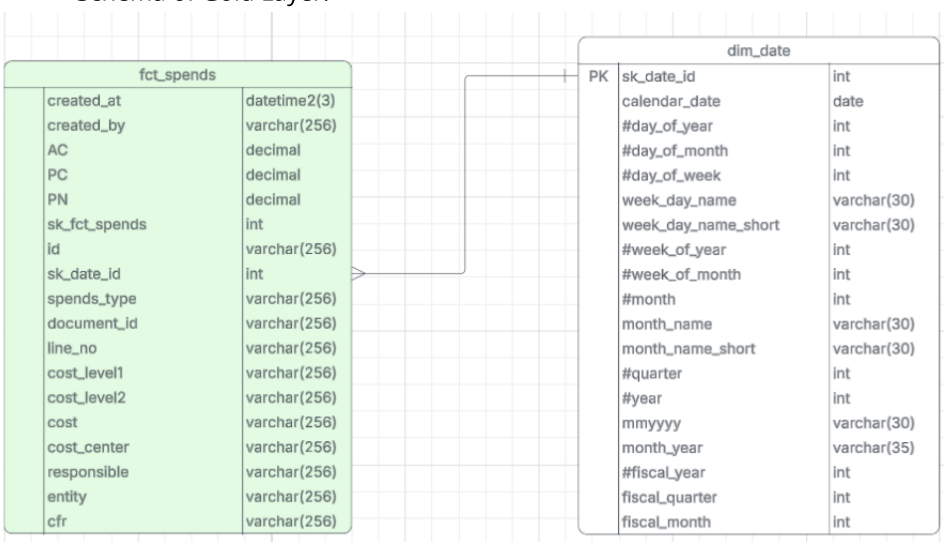
- system_name_schema_name.src_tbl_name
- Each table contains following technical fields:
    - TechExecutorRunID
    - TechProcessorRunID
    - TechProcessingDateTime
    - TechBusinessDateTime

The latest data snapshot is transferred into **Stage** layer of DWH as-is using pipeline with the DROP TABLE IF EXIST option

Data is then moved to the **Bronze** layer as-is through store procedure which accepts table name, and strategic of loading data (SCD1, SCD2, Full) as parameters.

The **Gold** layer contains single fact table with deformalized data based on the mentioned tables above and Calendar table. These table are linked by sk_date_id. Table name is FctSpends, which is fed by store procedure with Full Reload approach. Gold table contains surrogate key – **sk_fct_spends** and technical fields **created_at** and **created_by.** Since multiple source fact tables exist, the FctSpends table includes a key field named **SpendsType**, which indicates the originating source table.

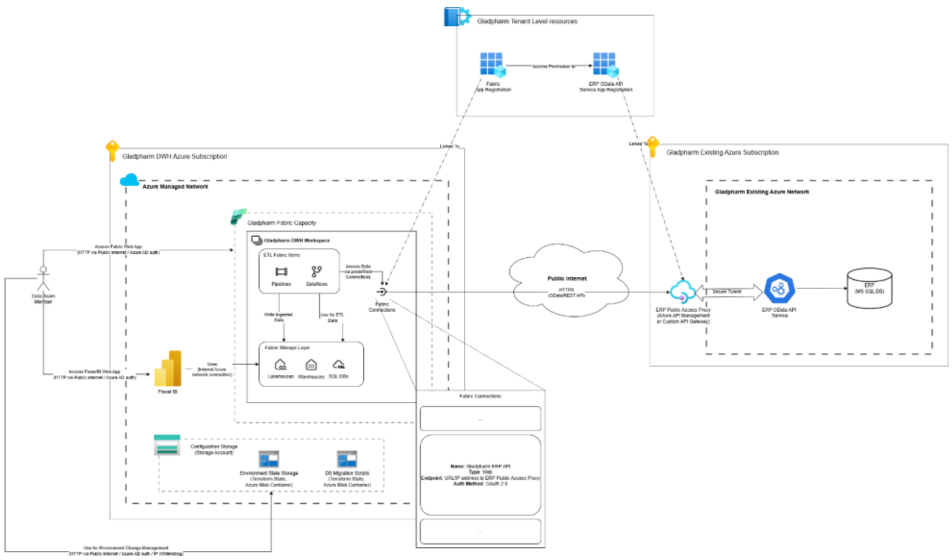
Schema of Gold Layer:



There is mapping of the columns, which describes how fact table is built:

Source Table.Column	Target Column	Comment
Сурогатний ключ (ID)	sk_fct_spends	
ОБ_АдмінВитрат_RecordType.Recorder	id	
Каз_БюдАдминВитрат_.Recorder	id	
ОБ_АдмінВитрат_RecordType.Period	sk_date_id	20260101
Каз_БюдАдминВитрат_.Period	sk_date_id	20250101
SourceFlag(Actual2025, Planning2025)	spends_type	будувати на основі Recorder_Type
ОБ_АдмінВитрат_RecordType.Документ_Key	document_id	
Каз_БюдАдминВитрат_.Документ_Key	document_id	
Каз_БюдАдминВитрат_.LineNumber	line_no	
Каз_БюдАдминВитрат_.LineNumber	line_no	
Статьи ОБ	cost_level1	
Статьи ОБ	cost_level2	
Статьи ОБ	cost	
Структура Предприятия.Description	cost_center	
Пользователию.Description	responsible	
Организация	entity	
Структура Предприятия.Description	cfr	
ОБ_АдмінВитрат_RecordType.Сума	PC,AC,PN (Total)	StandardODATA.Document_ЗаявкаНаРасходованиеДенежныхСредств *-1
Каз_БюдАдминВитрат_.Сума	PC,AC,PN (Total) *-1	

Applications Infrastructure/networking components



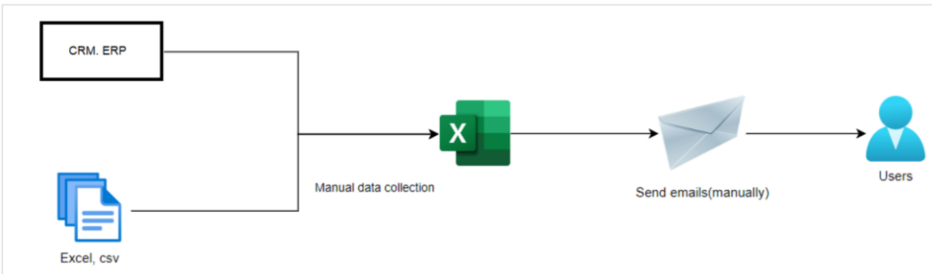
Building the POV solution we will rely on existing approach with providing the access to the customer data sources using API gateway with the secure (TLS, HTTPS) and Authenticated (OAuth 2.0) connections established via public internet.

Додано примітку [RK1]: Виправити зображення - замінити 1С на ERP

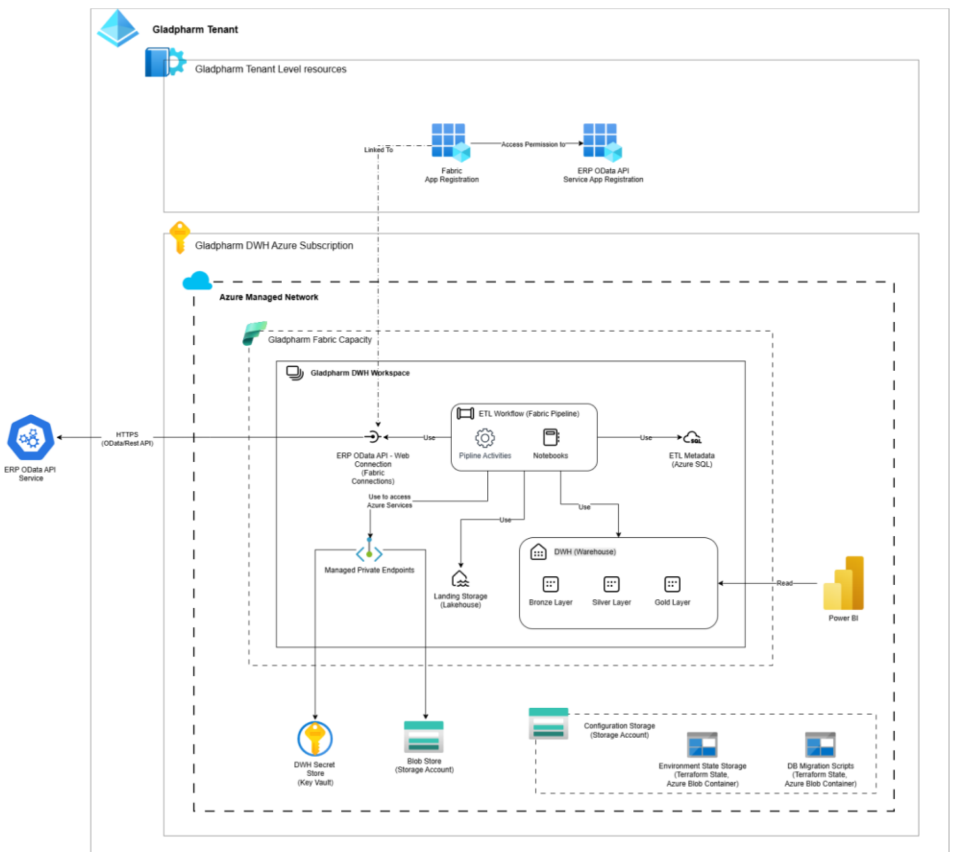
Visual representation of As IS vs Future Landscape

As IS Landscape

The data is analyzed and aggregated for further processing. After aggregation, it is enriched with proprietary information (such as project and division mappings). Reports are then prepared manually, and the results are distributed to users via email. Although a dedicated Data Warehouse and Analytics system exist within the client's landscape, the complexity of onboarding new processes through custom development prevents the process from being fully automated.



Future Landscape



The solution contains:

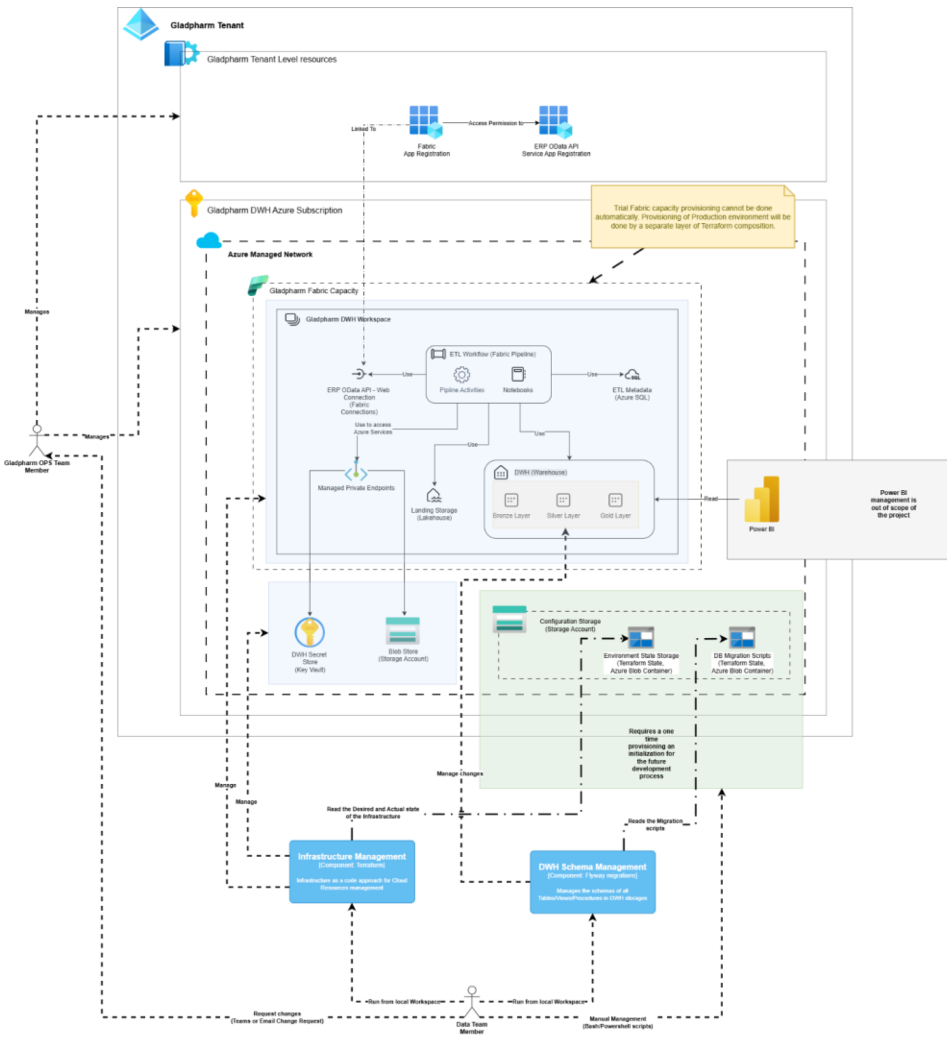
Додано примітку [RK2]: Виправити зображення вище. Замінити 1C на ERP

The proposed solution is based on Microsoft Fabric and deployed in the Azure. It connects to the on-premise data source from Gladpharm's environment using publicly exposed REST API managed by Azure API Management protected by OAuth 2.0 authentication layer. The data is ingested using OData protocol . The architecture ensures scalability, performance, and end-to-end security while enabling seamless integration between data sources, data storage, and analytical tools.

Azure DWH Environment

1. Gladpharm DWH Workspace: The main Microsoft Fabric Workspace that is a part of the Gladpharm Fabric Capacity. All the Fabric Items needed for ETL process and DWH storage are hosted in it, such as data pipelines, notebooks, lakehouses, and warehouses
2. Tennant level App registrations: The Azure App registrations and their configuration needed for establishing a secure and authorized connection between Fabric Activities and the Data Source's APIs.
3. Key Vault: Manages encryption keys, credentials, and connection strings used by pipelines or infrastructure provisioning tools.
4. Storage Account: Cloud-based storage for manual datasets, configuration files or other static assets required for ETL process.
5. Configuration Storage: Azure Storage account required for performing the environment configuration management done by infrastructure provisioning tools.

Provisioning and configuration management will be performed using Terraform HCL for infrastructure and Configuration management of DWH platform. The DB Schemas of DWH Storage Layer will be managed using Flyway migrations.



3. Data Processing and Storage Layers

- ETL Workflow: The Fabric Pipeline that performs an orchestration of the ETL flow. Contains Data Pipeline activities and Fabric Notebook calls required for performing all needed stages of the ETL Workflow.

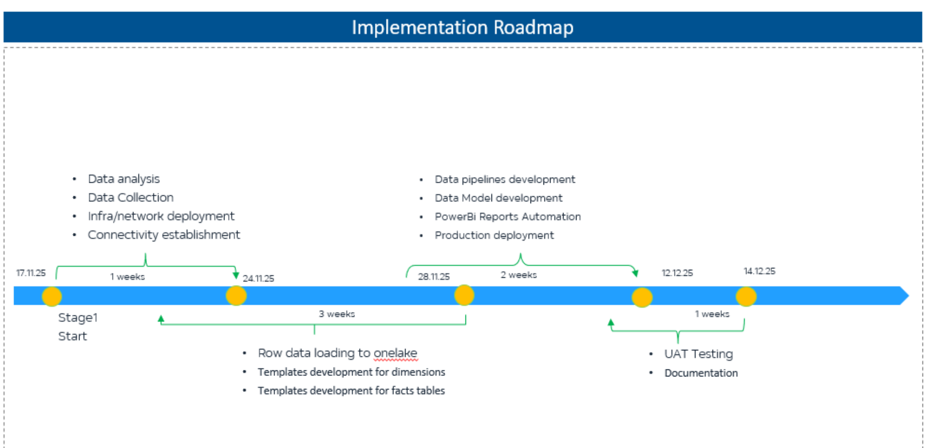
Додано примітку [RK3]: Змінити зображення вище. Замінити 1С на ERP

- Pipeline Activities: A set of Fabric pipeline's predefined actions that handle automated data ingestion, transformation, and loading (ETL/ELT) from MS SQL, ERP, and manual sources.
- Notebooks: Used for data actions that are not covered by Fabric Pipeline's activities or that required complex logic.
- ERP OData API Connection: preconfigured API client for the external ERP Data API Service. Required for Fabric Items to connect to external API services.
- Managed Private Endpoints: allows secure and private access to data sources or services that are not a part of the Fabric but managed by the Azure Platform itself.
- DWH: Serves as the central repository for raw and curated data, organized into the Bronze, Silver, and Gold layers:
    - Bronze Layer: Raw data ingested from source systems without transformations.
    - Silver Layer: Cleaned and standardized data, ready for analysis or integration.
    - Gold Layer: Aggregated and business-ready datasets used by reporting tools.
- Landing Storage: a storage for temp data ingested by the ETL Workflow before promotion it to the DWH. Required for protection of the DWH from the partial changes done by the failed ingestion processes.
- ETL Metadata: Store metrics produced by the ETL Workflow. Also stores other metadata required for it.

4. Reporting and Visualization
Microsoft Power BI: The primary visualization and reporting tool connected to the DWH, mainly to the Gold Layer.

Deployment Plan

Based on the project business goals, we recommend to implement solution within following phases:



Phases of PoC	Tasks	Start Date	End Date	Work Results
Stage 1 – Data Analysis & Infrastructure Setup	• Data analysis • Data collection • Infra & network deployment	03.11.25	14.11.25	• Full understanding of data sources • Connected environments • Ready infrastructure for next steps
Stage 2 – Model & Report Development	• Model development • BI report development	14.11.25	19.11.25	• First working version of the model • Draft BI report

	(non-automated) • Initial testing			• Verified logic & data structure
Stage 3 – Pipeline & Automation Development	• Data pipelines development • Modelling automation • Power BI report automation • Deployment to production environment	19.11.25	02.12.25	• Automated pipelines • Automated model refreshes • Automated BI report refresh • Deployed PoC solution
Stage 4 – UAT & Documentation	• UAT testing • Documentation	02.12.25	09.12.25	• Validated PoC by business users • Final PoC documentation & knowledge transfer

Phases of scaling	Tasks	Start Date	End Date	Work Results
Stage 1 – Architecture & Infrastructure Setup	• Fabric Capacity (F16/F32) provisioning • Dev/Test/Prod workspaces & RBAC • Network & Gateway configuration • Initial Lakehouse structure: Stage/Bronze/Silver/Gold • Security setup: encryption, access control, audit	**05.01.2026**	**19.01.2026**	• Ready and secured Fabric platform • All environments prepared for development
Stage 2 – Data Ingestion & Integration Setup	• Onboarding all data sources: MS SQL, Odoo CRM, Odoo Retail, Respos API, NIQ, Yieldigo • Building ingestion pipelines for each source • Establishing data retention rules • Manual file templating & validation (CSV/Excel)	**19.01.2026**	**09.02.2026**	• Unified ingestion layer • Standardized manual data upload process • Stable daily data collection
Stage 3 – Modelling, Automation & CI/CD	• Full Bronze → Silver → Gold transformations • Business models for all entities (≈1000 objects) • Data Quality Framework (schema checks, load completeness, alerts) • Pipeline orchestration & dependency management• CI/CD implementation (Dev → Test → Prod), Git versioning	**09.02.2026**	**05.03.2026**	• Automated data pipelines • End-to-end model automation • CI/CD ready development lifecycle • Data quality monitoring in place
Stage 4 – Reporting, Performance, Migration	• Migration of historical data (≈1 TB) • Migration of existing Power BI reports • Development of new reports & semantic models • Performance tuning (EoM/EoQ/EoY scenarios) • SLA verification & load optimization	**05.03.2026**	**25.03.2026**	• Full historical data in Fabric • Optimized reporting environment • Confirmed performance vs SLA

Stage 5 – UAT, Documentation & Go-Live	• User acceptance testing • Bug fixing & adjustments • Operational documentation & handover • Go-Live & post-launch support	**25.03.2026**	**05.04.2026**	• Production-ready solution • Team fully onboarded • Completed full implementation

Budget

Based on a questionnaire sessions we collected the next metrics that will be used for consumption estimations:

System	Access Aproach	Access Protocol	Projected number of records per year
ERP	API Server	Odata	2 000 000
Planning and Budgeting	API Server	Odata	1 000 000
Treasury	API Server	Odata	1 000 000
Accounting System	API Server	Odata	1 000 000
Transport Management System	API Server	Odata	1 000 000
MS SQL	Internal Network or API Server	MS SQL	38 000 000
Sales Operations	Internal Network or API Server	MS SQL	30 000 000
Visit Activity	Internal Network or API Server	MS SQL	5 000 000
Market Analytics	Internal Network or API Server	MS SQL	3 000 000
Other		Excel,Dataverse, Azure DB, Share Point, Rest API	2 500 000
Project Activity	*Excel*	Excel	1 000 000
	Rest API	Share Point	
	OData Web API	Dataverse	
Document Management	Rest API	Rest API	500 000
Vehicle Tracking (GPS)	Rest API	Rest API	1 000 000
			44 500 000

Total historical data volume (actual) - preliminary 500GB-1TB. Using the Fabric Calculator guide of "If you only have the logical (un-compressed value) divide it by 6.", we define the value of 160GB for calculator.
We were not able to collect the total number of objects in the context of the project which is the required metric for the Fabric Capacity Estimator. Based on the collected metrics and our previous experience we made the following assumptions.

- ERP System - 2M records/year
  Typical ERP structure: Master data, transactional data, configuration
  Record-to-table ratio: ~10,000-15,000 records per table average
  Estimated: 130-200 tables

- Accounting System - 1M records/year
  Typical accounting system: Chart of accounts, journal entries, reconciliations
  Record-to-table ratio: ~8,000-12,000 records per table
  Estimated: 80-120 tables
- Transport Management System - 1M records/year
  Typical fleet system: Vehicles, drivers, routes, fuel, maintenance
  Record-to-table ratio: ~15,000-20,000 records per table
  Estimated: 50-70 tables
- MS SQL - 38M records/year, Highest volume
  Typical data: Accounts, contacts, opportunities, activities, leads, campaigns
  Record-to-table ratio: ~150,000-250,000 records per table (high transactional volume)
  Estimated: 150-200 tables
- Project Activity - 1M records/year
  Excel, SharePoint, Dataverse combined
  Estimated: 15-25 tables (relatively simple structures)
- Document Management - 500K records/year
  Typical DMS: Documents, versions, workflows, approvals
  Estimated: 10-20 tables
- Vehicle Tracking - 1M records/year
  Typical GPS tracking: Location points, trips, geofences, events
  Estimated: 5-15 tables

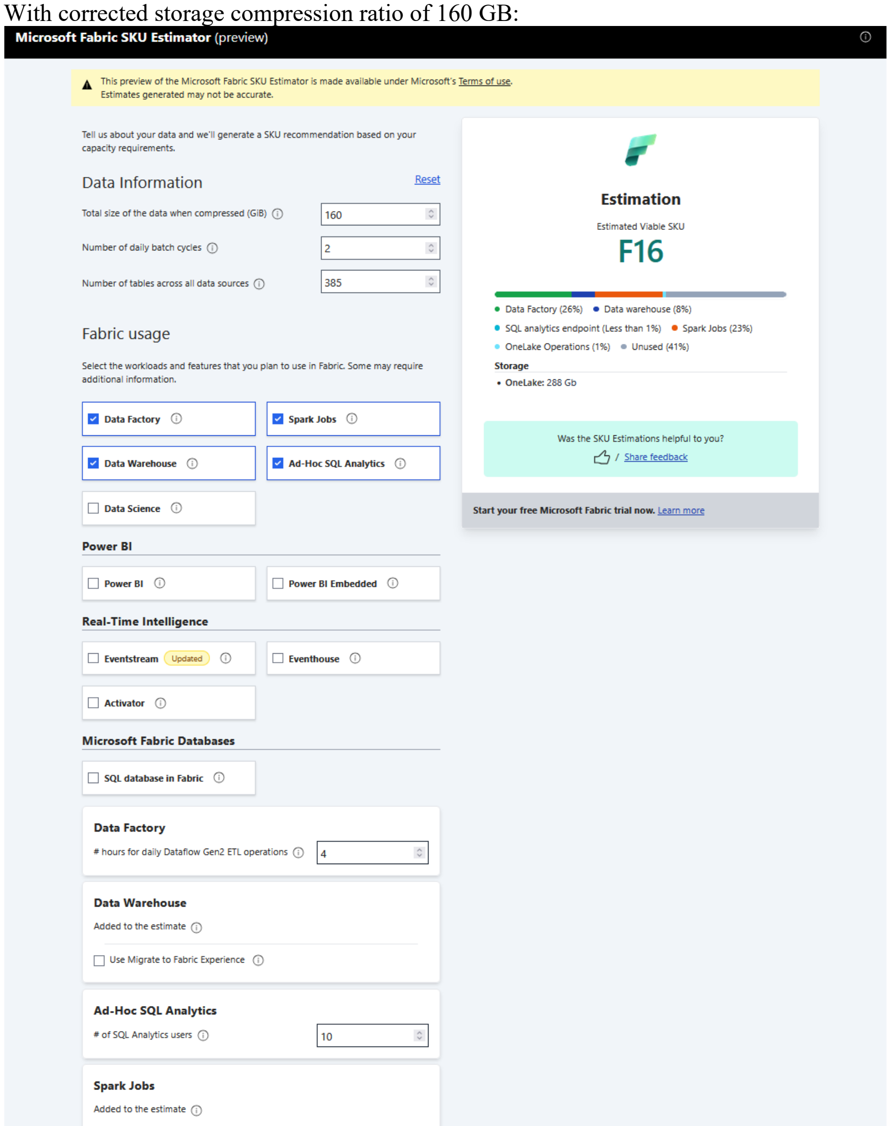
Total Estimation: 440-650 tables
Most Realistic Range: 500-550 tables

Relevant for DWH: 60-70% of source tables (many technical/system tables won't be needed) - 300-385 tables

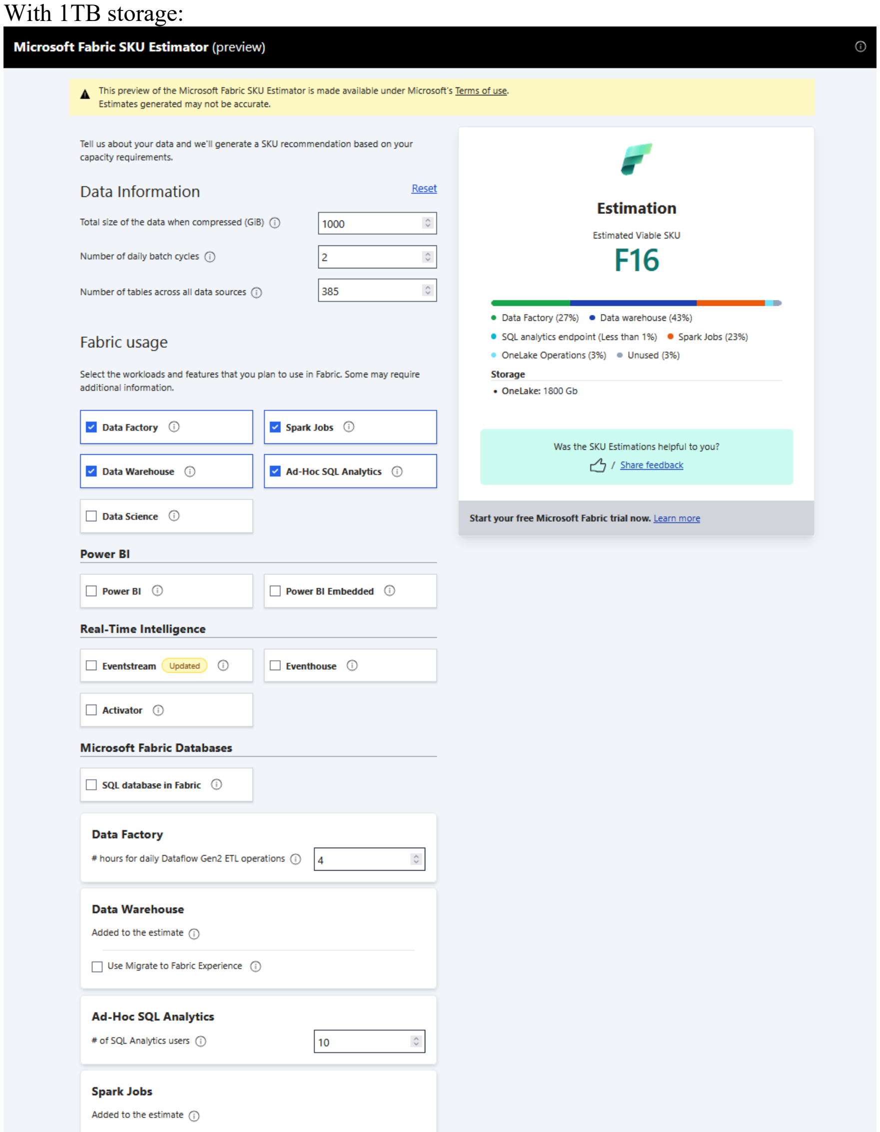
For POV we will use ERP System and Project Activity: 145-225 in total, Relevant for DWH 100-160
Microsoft Fabric Capacity Estimator:

Main scope

With corrected storage compression ratio of 160 GB:



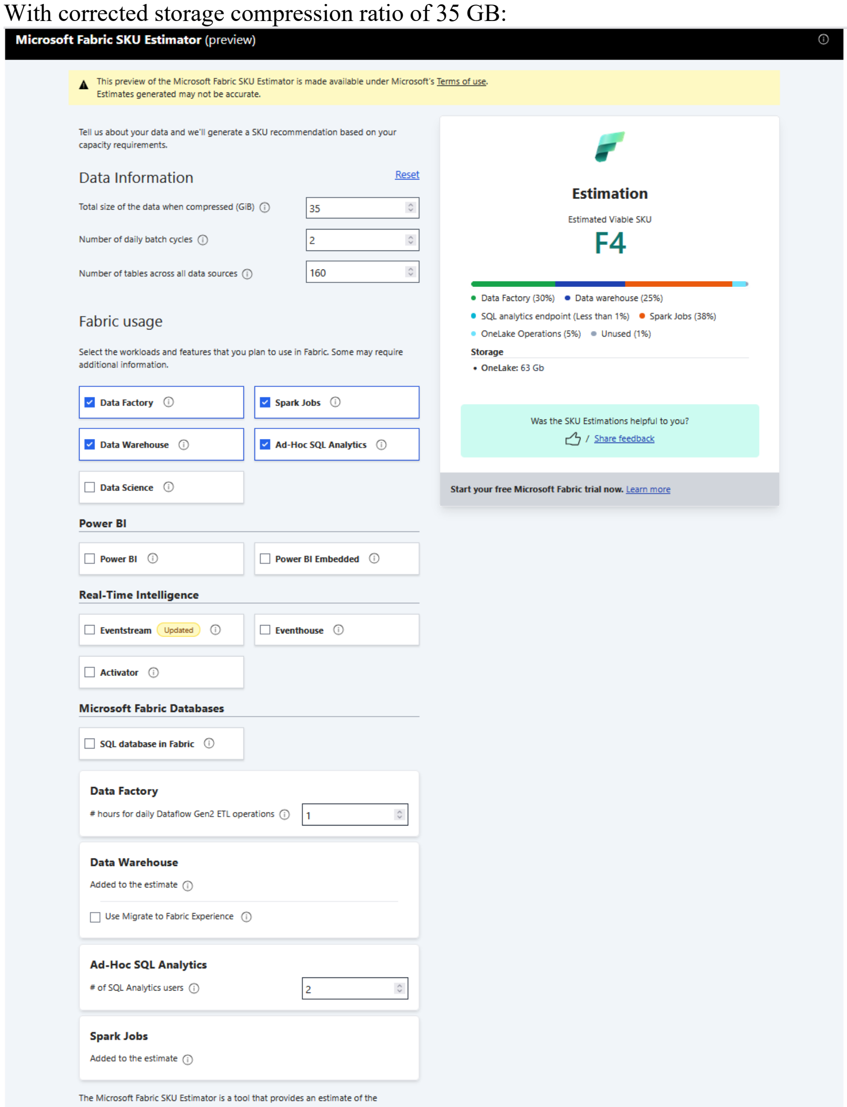
With 1TB storage:



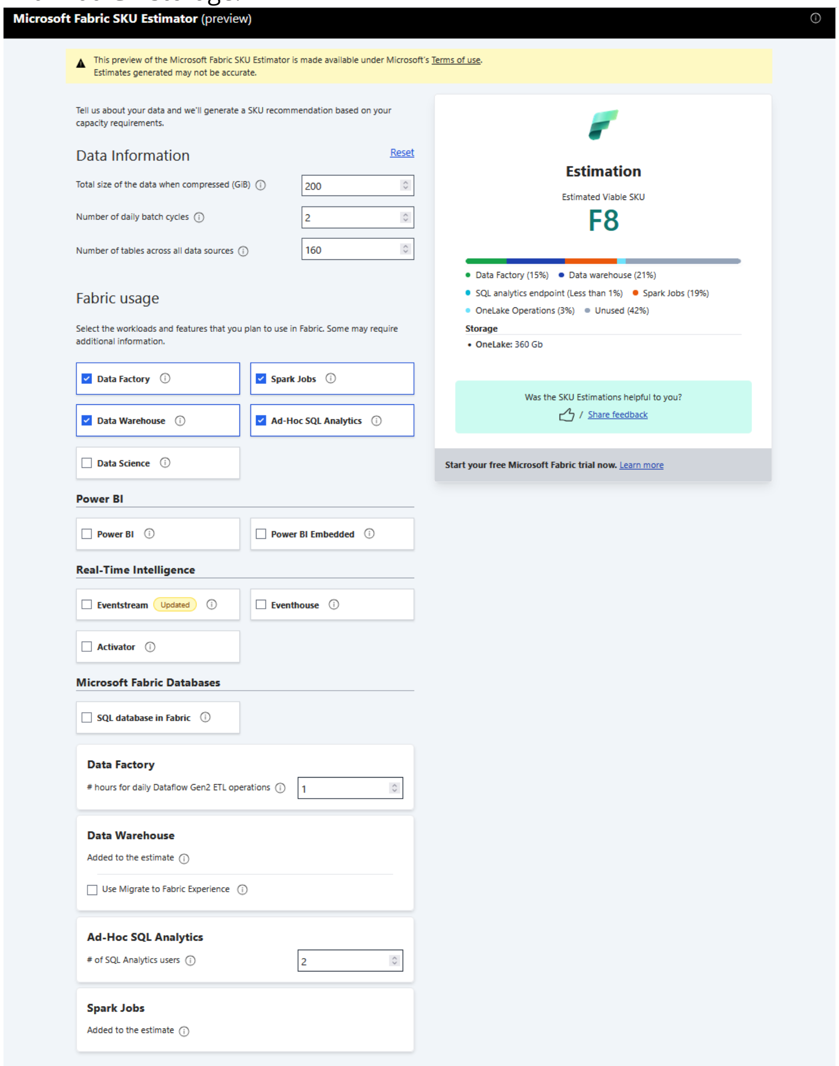
POV scope

For POV scope with ERP and Project Activity systems in the scope, let’s assume their total storage footprint is 20% of the all storage consumption. Which is 200 GB and corrected value of 35GB.

With corrected storage compression ratio of 35 GB:



With 200 GB storage:



Planned Azure Consumption:

Azure calculator (North Europe is selected as cheaper region than Poland)		
Main scope	MS Fabric, F16	Calculation Estimated monthly cost: $2,260.60 With 1 year reserved: $1,361.40
Main scope	MS Fabric, F64 with access to free read-only PowerBi licenses that can free 100 Power Bi Licenses (~$1400)	Calculation Estimated monthly cost: $8,876.80 With 1 year reserved: $5,280.00
POV scope	F4	Calculation Estimated monthly cost: $555.60 With 1 year reserved: $330.81
POV scope	F8	Calculation Estimated monthly cost: $1,117.88 With 1 year reserved: $668.28